

Rapid Revision Summary

Last-minute review · condensed from 18 papers and mark schemes, Oct 2020 – Jun 2026

THE PAPER

1 h 30 min · 75 marks · 8–11 questions · ≈ 1.2 min per mark.
Every paper: binomial, vectors, integration, parametrics (18/18); implicit, DEs, volumes (17); proof by contradiction (15).

TOP 12 MARK-LOSERS

- Calculator-banned questions: no working = **no marks**.
- Contradiction proof: **assumption in words**, contradiction stated, conclusion.
- Skew = **do not intersect** (3rd equation fails) **and not parallel**.
- Angles in degrees when asked; vectors from the same vertex.
- No mixed notation (5i, 6j, -2k) or coordinates for a vector.
- Binomial: take out a^n ; bracket (bx/a); state $|x| < \dots$.
- “+–” for “–” in a coefficient = A0.
- Substitution: change dx **and limits**; u = x not allowed.
- Exact: 2 ln 2, not 1.386. Keep π , ln, surds.
- + c, then use the condition. Never just check a given answer.
- Connected rates: substitute h or r, **not t**.
- Normal/tangent in integers with “= 0”.

PROOF BY CONTRADICTION

FOUR MARKS

Assume opposite (words) → use it → **contradiction** stated → **conclude**.

- n^3 even $\Rightarrow$ n even: assume $n = 2p + 1 \Rightarrow n^3 = 2(4p^3 + 6p^2 + 3p) + 1$ odd. Contradiction.
- $\sqrt[3]{2}$ irrational: $\sqrt[3]{2} = a/b$ lowest terms $\Rightarrow a^3 = 2b^3 \Rightarrow a$ even $\Rightarrow b^3 = 4k^3 \Rightarrow b$ even. Common factor.
- Curves don't meet: $x^4 + 8x^2 + 15 = 0$ impossible ($x^4, x^2 \geq 0$).
- No stationary point: $\sin x = 2 + 3x^2 \geq 2$, but $|\sin x| \leq 1$.
- Not geometric: $(1 + 2k)^2 = k(3 + 3k) \Rightarrow k^2 + k + 1 = 0$, discriminant < 0 .
- Complete every case ($3k + 1$ **and** $3k + 2$).

PARTIAL FRACTIONS

Denominator	Form
$(ax + b)(cx + d)$	$A/(ax + b) + B/(cx + d)$
$(ax + b)(cx + d)^2$	$A/(ax + b) + B/(cx + d) + C/(cx + d)^2$
improper	divide first

- $(6x^2 - 16)/((x + 1)(2x - 3)) = 3 + 2/(x + 1) - 1/(2x - 3) \Rightarrow \int = 3x + 2 \ln|x + 1| - \frac{1}{2} \ln|2x - 3| + c$.

- Cover-up for roots, compare coefficients for the repeated term.

BINOMIAL EXPANSION

FORMULA

$$(1 + x)^n = 1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \dots, |x| < 1$$

$$(a + bx)^n = a^n(1 + bx/a)^n, \text{ valid } |x| < |a/b|$$

- $(1 - 4x)^{-3} = 1 + 12x + 96x^2 + 640x^3, |x| < \frac{1}{4}$.
- $(2 + 5x)^{-2} = \frac{1}{4} - (5/4)x + (75/16)x^2 - (125/8)x^3, |x| < 2/5$.
- Two expansions: use the **narrower** validity.
- Unknown n, k: match coefficients; reject excluded values.
- Approximation: substitute x into the series, show it.

IMPLICIT DIFFERENTIATION

RULES

$$d/dx f(y) = f'(y) y' \cdot d/dx(xy) = y + xy' \cdot d/dx(a^y) = a^y \ln a \cdot y' \cdot \text{constants} \rightarrow 0$$

- $xy^2 = x^2y + 6$ at (2, 3): $y' = (2xy - y^2)/(2xy - x^2) = 3/8 \Rightarrow 3x - 8y + 18 = 0$.
- Turning point: numerator = 0 → substitute into the curve.
- Max/min x: denominator = 0 ($dx/dy = 0$).
- $y = x^{\sin x}$: $\ln y = \sin x \ln x \Rightarrow y'/y = \cos x \ln x + \sin x/x$.

PARAMETRIC EQUATIONS

TOOLKIT

$$dy/dx = (dy/dt) \div (dx/dt)$$

$$\text{Area} = \int y (dx/dt) dt \cdot \text{Volume} = \pi \int y^2 (dx/dt) dt \text{ (t-limits)}$$

$$\text{Cartesian: eliminate } t \text{ (} t = \dots, \text{ or } \sin^2 + \cos^2 = 1, \cos 2t = 1 - 2\sin^2 t, \sec^2 = 1 + \tan^2)$$

$$\text{Domain/range from the t-interval.}$$

- Tangent meets curve again: substitute $x(t)$, $y(t)$ into the line; the tangent point is a repeated root: $(t + 2)^2(7t - 20) = 0$.

INTEGRATION

BY PARTS

$$\int u v' dx = uv - \int v u' dx. u = \ln x \text{ if present; else } u = \text{polynomial.}$$

$$\int \ln x/x^2 dx = -\ln x/x - 1/x + c \cdot \int x \cos 3x dx = (x/3) \sin 3x + (1/9) \cos 3x + c$$

$$x^2 e^x: \text{twice. } e^x \cos 2x: \text{twice, then solve for I.}$$

BY SUBSTITUTION

Change integrand, $dx \rightarrow du$, limits. Don't go back to x for definite integrals.

$x = 2 \sin u$: use $\cos^2 = 1 - \sin^2$, $u = e^x + 4$: $dx = du/(u - 4)$.

- $\int f'/f = \ln|f|$; $\int f' f^n = f^{n+1}/(n+1)$.
- $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$; $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$; $\sin x \cos x = \frac{1}{2} \sin 2x$.
- Combine logs to the requested form ($p + q \ln 2, \ln(a/b)$).

VOLUMES**VOLUME**

$V = \pi \int y^2 dx$. $y = e^{0.5x} - 2$, $0 \rightarrow 2 \ln 2$: $\pi \int (e^x - 4e^{0.5x} + 4) dx = \pi(8 \ln 2 - 5)$.

- Square y fully. Divide/partial fractions first if needed.
- Region with a line: add/subtract cone, cylinder or triangle.
- Context: units; density = mass $\div$ volume.

DIFFERENTIAL EQUATIONS**SEPARATE**

$\int 1/g(y) dy = \int f(x) dx + c \rightarrow$ use condition $\rightarrow$ rearrange.
 "Rate of decrease $\propto 1/r^2$ "; $dr/dt = -k/r^2$.

- Tank (Jan 21): $dH/dt = -(H-5)(H+3)/40 \Rightarrow (H-5)/(H+3) = \frac{1}{2}e^{-0.2t} \Rightarrow H = (10 + 3e^{-0.2t})/(2 - e^{-0.2t})$; $H = 8$ at $t = 5 \ln(11/6) = 3.03$; $H \rightarrow 5$.
- Long term: $t \rightarrow \infty$. "Ever full?": compare limit with capacity.
- Tank inflow/outflow: $A dh/dt = \text{rate in} - \text{rate out}$.

CONNECTED RATES

- $dh/dt = (dV/dt) \div (dV/dh)$.
- Cone (Oct 24): $r = 2h/5$, $V = 4\pi h^3/75$; $t = 90 \text{ s} \Rightarrow V = 180\pi \Rightarrow h = 15 \Rightarrow dh/dt = 1/18 \text{ cm s}^{-1}$.
- Sphere: $dV/dr = 4\pi r^2$; $70\pi \div 100\pi = 0.7 \text{ cm s}^{-1}$ at $r = 5$.

VECTORS**TOOLKIT**

Line $r = a + \lambda d$: unit vector $d/|d|$; $|v| = \sqrt{x^2 + y^2 + z^2}$

$\cos \theta = a \cdot b / (|a||b|)$; acute angle between lines: $|a \cdot b|$

Foot of perpendicular from P : $X(\lambda)$, $PX \cdot d = 0$

Intersect: solve 2 equations, check the 3rd

Reflection: $OB = 2OX - OA$; Area = $\frac{1}{2}|a||b| \sin \theta$

SKEW (3-4)

Solve two equations $\rightarrow$ third fails $\rightarrow$ do not intersect.

Directions not multiples $\rightarrow$ not parallel. Hence skew.

- Angle ABC: use $BA \cdot BC$ (-26), not $AB \cdot BC$: $B = 112.65^\circ$.
- Nearest point to O on $(1, -10, -9) + \lambda(4, 4, 2)$: $\lambda = 1.5$, $X(7, -4, -6)$.
- $AB : AC = 1 : 3 \Rightarrow c = 3b - 2a$.
- Fixed distance from a point: quadratic in $\lambda \rightarrow$ two points.

FINAL 60 SECONDS

- Exact form? Degrees? Units? " $= 0$ " seen?
- Every proof: assume $\rightarrow$ contradict $\rightarrow$ conclude. Every skew: not meet + not parallel.
- Every integral: $+ c$ or exact limits; every DE: condition used.